

Tapan Sethi

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EDUCATION

Indian Institute of Technology, Guwahati

Assam, India

Bachelor of Technology in Computer Science and Engineering

Nov 2020 – July 2024

- CGPA: 7.24
- Core CS: Database Management Systems, Software Engineering, Design and Analysis of Algorithms, Artificial Intelligence, Deep Learning
- Basic Mathematics: Probability Theory & Random Processes, Elementary Number Theory & Linear Algebra, Optimization

EXPERIENCE

Research Engineer

Aug 2024 – Present

Centre for Development in Telematics

New Delhi, India

- Engineered and optimized core backend modules within a production Kubernetes-based IoT platform processing ~6M requests/day (70 avg RPS, 110 peak RPS) across 5k+ devices.
- Scaled system infrastructure from 2k to 3.5k daily active devices (75% growth) while maintaining <0.001% server error rate through traffic analysis and ELK-based failure monitoring.
- Led end-to-end development of a historical data tracking feature for Indian Railways' real-time wagon monitoring on a 20+ pod architecture with application-specific, horizontally scaled services for differentiated throughput.
- Replaced the inefficient PSK-based communication with a PKI-driven REST client, reducing socket-level failures and accelerating resource load time for 400+ devices per subscriber.
- Architected and delivered a regulatory-compliant delete API with scoped access controls, deployed in 48 hours for mission-critical clients (e.g., CRIS), addressing a critical limitation in the oneM2M standard.
- Diagnosed and resolved a protocol-level failure during field deployment, restoring full system functionality for 50+ active vendors within minutes, enabling uninterrupted validation of IoT devices across the network.
- Orchestrated the containerization of a monolithic Wildfly module into a standalone Dockerized microservice, decoupling the database initialization for faster system bootstrapping.
- Integrated a multi-tenant License Manager client, implemented automated license file injection via ConfigMaps and Volumes and Java bytecode obfuscation to secure platform IP.

PROJECTS

Compiler for nano-C (subset of C), Course Project | GitHub

Jan 2023 – May 2023

- Designed a compiler for the nano-C language, covering all four phases of compiler design.
- Built a lexical analyzer using FLEX (Phase 1).
- Created a syntax analyzer using Bison for generating productions (Phase 2).
- Generated machine-independent code using syntax-directed translation (Phase 3).
- Produced target code via simple table-lookup-based code generation (Phase 4).

IoT Notification Handler | GitHub | Live

Nov 2025 – Dec 2025

- Built a Python service using Flask to receive IoT sensor data securely.
- Parsed JSON payloads and stored notifications in Neon PostgreSQL.
- Containerized with Docker and deployed on Render with HTTPS for secure cloud access.
- Processed live notifications for smart agriculture applications, supporting real-time decision-making.

TECHNICAL SKILLS

- Languages: Java (Core & Jakarta EE), C/C++
- Frameworks: JPA, JDBC, JUnit, REST, XML Schema Definitions (XSD), JSON
- Cloud & Databases: Docker, Kubernetes (K8s), PostgreSQL
- Developer Tools: GitLab, VS Code, Eclipse, pgAdmin4, Postman, MQTTX, ProGuard

ACHIEVEMENTS

- Achieved global ranks of 476, 1061 in contests with 42k, 27k participants respectively and rated Specialist on Codeforces (Profile).
- Qualified for the Indian National Mathematics Olympiad (INMO) 2020, one of only 2 out of over 900 participants.
- Authored 4 technical proposals officially Agreed to the oneM2M Global Standards.
- Selected as an Edison Engineering Development Program Intern at GE Healthcare, working on a logger management system for X-ray logs using Fluentd.